ME 144L LE 2: Physical Signal Processing and Analysis

Memorandum

Date: 2/7/2026
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Digital Filter Results

Given the data from the lab exercise, and following similar code structures to that of the pre-lab, the provided array of values can be modeled as a signal over time. Provided that the sampling rate is 64 Hz, and there are 64 items in the array, it can be determined that the total period of this sample is 1 second. From this, we can construct a spectral plot to analyze the input data.

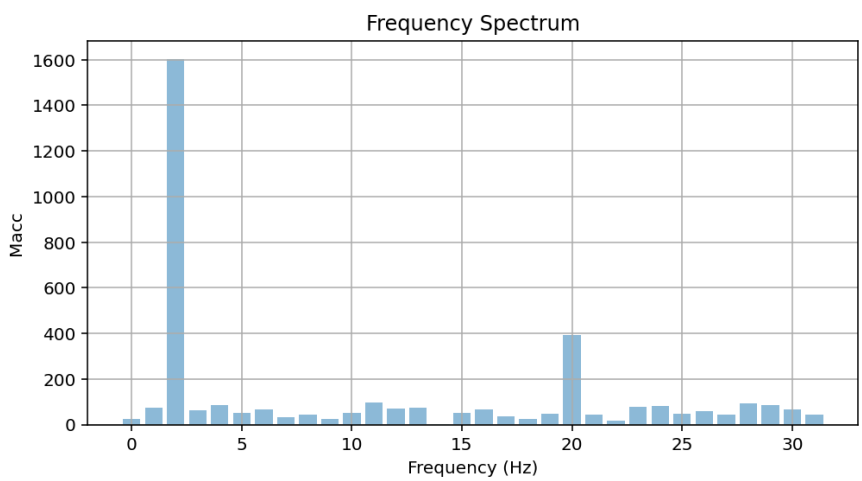

Figure I: Spectral Plot of Given Array in Lab Exercise 4

Using the same processes of creating a discretized low-pass RC filter from the pre lab, the same solution can be applied create a filtered function. After tuning the capacitance value to optimize filtering for roughly estimating input data while holding R constant, the best determined value came to C = 0.05. This C value paired with an R = 1, the cutoff frequency is roughly 3.183 Hz. Below is the raw array from exercise 4, shown alongside the digital RC filter with $\tau_t = 0.05$.

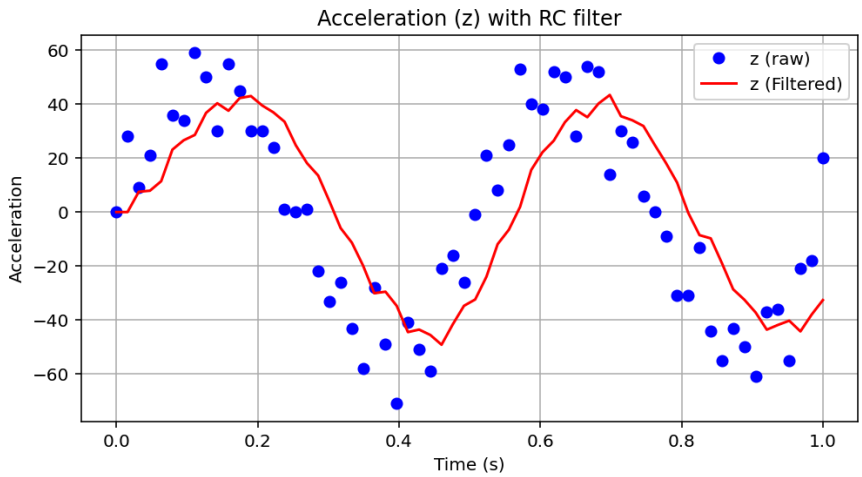

Figure II: Sample Acceleration (rad/s^2) vs. Time (s) with low-pass RC filter

<u>Flywheel Spin Down Data</u>

Using the same RC filter controls from the lab experiment, the code can be easily adapted to take the required '.csv' files and given rotational inertia values to create a new flywheel simulation. From this simulation, the B value can be estimated for each.

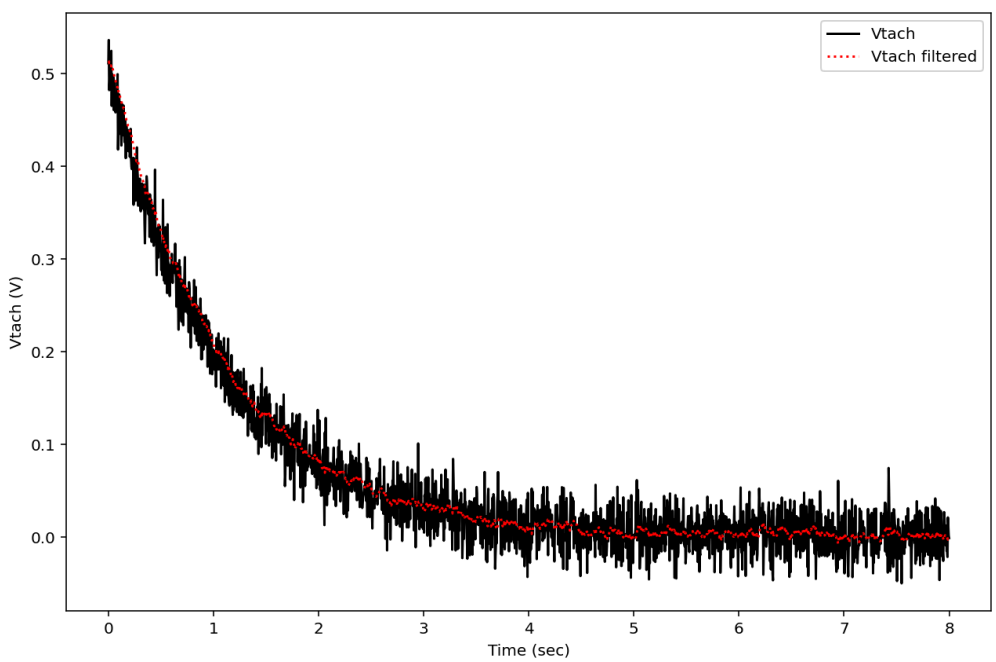

Figure III: Case 1 Tachometer Voltage (V) vs. Time (sec), B = 14.745

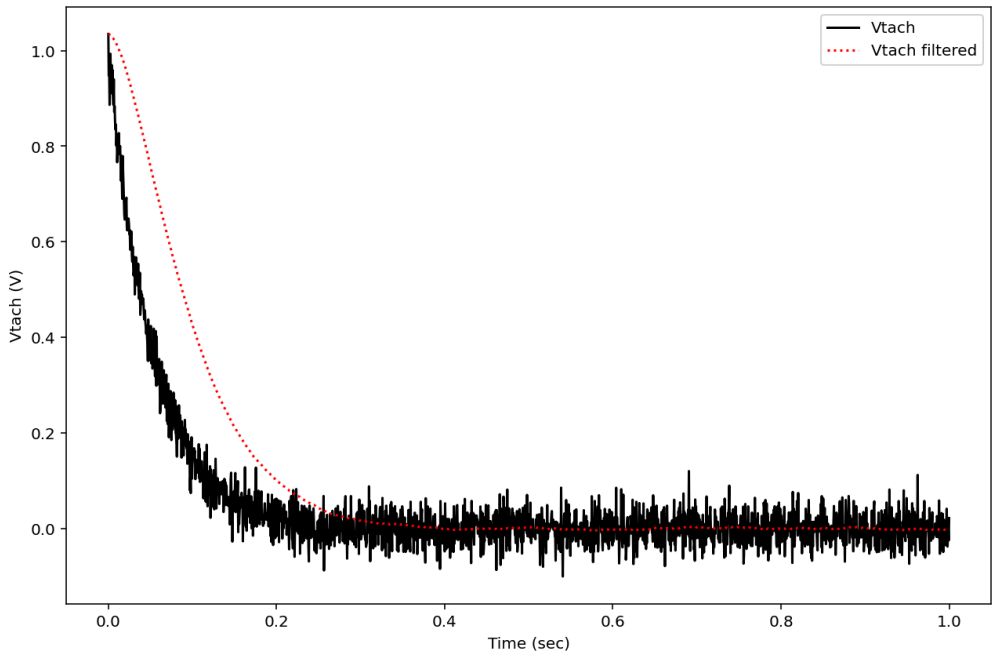

Figure IV: Case 2 Tachometer Voltage (V) vs. Time (sec), B = 1.690

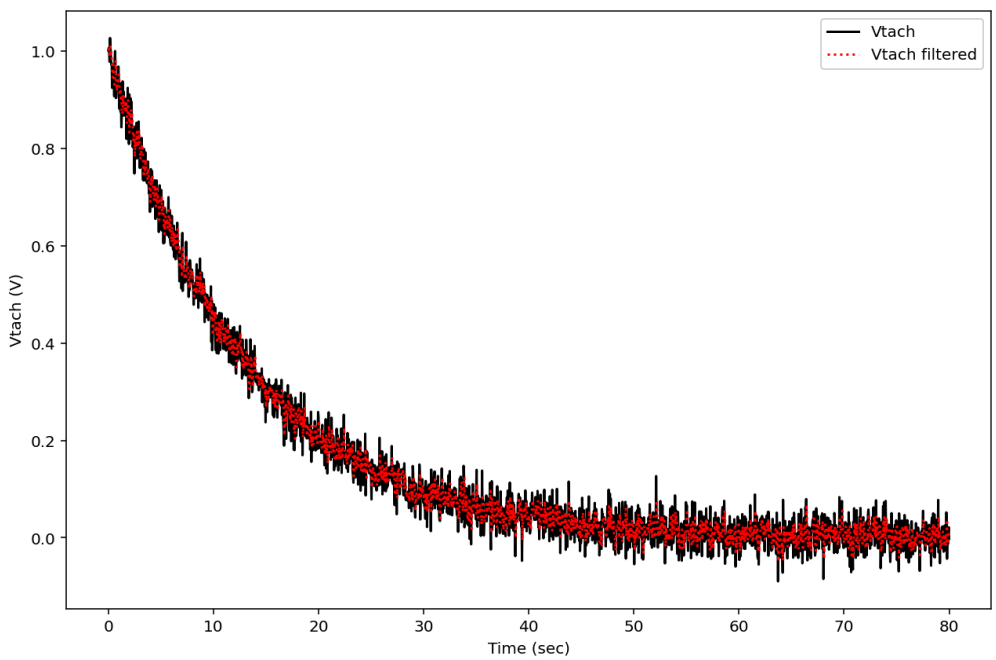

Figure IV: Case 3 Tachometer Voltage (V) vs. Time (sec), B = 48.73

<u>Summary of Results</u>

The filter controls were consistent with those used in the pre-lab acceleration analysis, utilizing a time constant of $\tau_t = 0.01$ s, derived from C = 0.01 and R = 1. This filtering process smoothed the raw voltage data (Vtach) into a cleaner signal (Vtach filtered), which was then compared against the theoretical decay models to estimate the damping coefficient B for each case.

Case Number	Filter Time Constant, τ_t (s)	Rotational Inertia, J (kg*m^2)	Estimated Damping Coefficient, B (Nms/rad)
Case 1	0.05	16.22	14.745
Case 2	0.05	0.0634	1.690
Case 3	0.05	633.58	48.736

Figure V: Summary of Tachometer Voltage Data for Various Flywheels

Note that the filter time constant τ_t of 0.05 s was maintained from the initial calibration described in the acceleration analysis from the lab exercise 4. The B values are taken directly from the curve-fit results presented in the figure captions.